Tender Submission:

AlSahra Switchgear LLC

Medium-Voltage Metal-Enclosed Switchgear — 13.8 kV

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Executive Summary

AlSahra Switchgear LLC proposes the ZX2-Plus indoor, metal-clad, withdrawable vacuum circuit breaker (VCB) switchgear for 13.8 kV, 60 Hz applications in mineral processing facilities. The solution prioritizes personnel safety with internal arc classification to AFLR at 31.5 kA for 1 s and integrates numerical protection relays with IEC 61850 Edition 2 capabilities. The product line is well-proven in the region and is intended to be housed in a ventilated electrical room designed for high ambient conditions typical to the Kingdom of Saudi Arabia.

Where the Company Standard ES-EL-013.8KV-001 specifies environmental and enclosure performance that exceed our baseline configuration, we outline feasible upgrade paths. Specifically, upgrades are available for ingress protection (IP4X external) and enhanced thermal capability for sustained operation at elevated room temperatures. These options may influence cost and schedule and are typically confirmed during post-award engineering.

[Ref §3] Service Conditions

The ZX2-Plus standard configuration is qualified for operation at an ambient of +40 C. The Company Standard calls for operation up to +55 C, with a continuous room design temperature of +50 C. We acknowledge this gap. AlSahra offers two mitigation approaches: (a) thermal derating of feeder and busbar currents in accordance with manufacturer curves, or (b) deployment of the high-temperature variant with upgraded ventilation and component insulation ratings. Humidity up to 95 percent (non-condensing) and altitude up to 1000 m are acceptable in both variants.

[Ref §4] Electrical Ratings

The offered equipment is designed for the 17.5 kV class for 13.8 kV service. Insulation performance satisfies LI (BIL) 95 kV and AC withstand 38 kV for 1 minute. Short-circuit withstand is 31.5 kA for 1 s for incomers and bus sections. The base busbar rating is 2000 A continuous in standard ambient; increased currents may be feasible with the high-temperature variant or with reduced temperature rise targets at the specified operating environment.

[Ref §5] Internal Arc and Personnel Protection

ZX2-Plus has been type-tested for internal arc classification IAC AFLR at 31.5 kA for 1 s per IEC 62271-200. Operator-side protection is achieved with reinforced doors and pressure relief flaps. For pressure exhaust management, the conceptual arrangement assumes top-mounted relief connected to a duct. Final routing, cross-sectional sizing, and termination to a safe area are to be defined during detailed design with the building engineer and the Owner's representative.

[Ref §6] Ingress Protection and Compartmentation

The baseline enclosure provides IP3X at the front external surfaces and IP2X between internal compartments. The Company Standard requires IP4X external. AlSahra can supply a gasket and filter kit, along with verified test documentation, to achieve IP4X external. This upgrade path is commonly selected for dusty environments but may increase maintenance due to filter loading in desert conditions.

[Ref §8] Secondary Systems and Protection

Protection and control are implemented using numerical relays that support IEC 61850 Edition 2, including GOOSE messaging for interlocking and trip signals where appropriate. Redundant Ethernet paths are supported via dual network interfaces. Disturbance records are available in COMTRADE format, and time synchronization via SNTP is standard. PTP can be provided when required by the Substation Automation System.

[Ref §9] Earthing and Bonding

A copper earth bar runs the full length of the switchgear lineup, with bonding of doors, instrument transformer cases, and withdrawable elements. The target substation earth resistance of 1 ohm or better is project-specific; AlSahra will connect to the building earth grid in accordance with the construction drawings.

[Ref §10] Installation Environment and Seismic

Clearances, access envelopes, and ventilation openings are indicated on the general arrangement drawings. AlSahra anticipates collaboration with the building designer to confirm airflow rates for a room temperature of +50 C. Seismic anchorage design to the Saudi Building Code is by others; anchor bolt layouts will be issued once the foundation details are available. If required, AlSahra can provide reaction loads for structural checks.

[Ref §11] Quality, Coatings, and Testing

A project-specific Inspection and Test Plan (ITP) will define hold and witness points for Factory Acceptance Testing (FAT) and Site Acceptance Testing (SAT). Routine tests include high-voltage withstand, insulation resistance, functional checks, and point-to-point wiring verification. Where IEC 61850 is in scope, functional tests, GOOSE subscriptions, and time synchronization will be demonstrated. Coating systems for enclosures can be supplied to meet ISO 12944 C4/C5.

[Ref §12] Documentation Deliverables

Deliverables include general arrangement drawings, single-line diagrams, wiring schematics, protection and coordination study reports with CT saturation checks, arc-flash assessment and label set, data sheets for CTs and VTs, relay ICD/CID files, FAT and SAT protocols and reports, operation and maintenance manuals, and recommended spares.

Appendices — Evidence Excerpts

Datasheet highlights: "+40 C (standard)", "IP3X (front)", "2000 A continuous".

GA notes: "Pressure relief ducts: to be defined by site", "Seismic anchorage: by others".

Document Register

- DRW-EL-SWGR-GA-0001 — Switchgear General Arrangement — Rev A — PDF
- MDS-EL-VCB-ZX2 — VCB and Panel Datasheet — Rev B — PDF
- CAL-EL-PROT-COORD — Protection and Coordination Study — Rev 0 — PDF
- ITP-EL-SWGR-FAT — FAT/SAT Inspection and Test Plan — Rev 1 — PDF
- REP-EL-ARC-FLASH — Arc-Flash Hazard Assessment — Rev 0 — PDF